

Causal Mechanisms of Activation-Monitor Evasion in a Neural Chameleon

Author(s) TBD

Private manuscript candidate – not released or submitted

Abstract

Activation monitoring aims to detect safety-relevant information in a model’s hidden states, but a learned model may alter those states when it knows which concept is being monitored. We investigate the causal implementation of this behavior in one released Gemma-2-9B Neural Chameleon checkpoint. Using paired, teacher-forced natural activation transplants, we select a 16-component population using only four benign discovery concepts, freeze it, and test it without modification on held-out benign concepts and a locked safety set. On deception, the population rescues 0.947 of the trigger-induced probe-suppression gap and induces 0.987; on harmfulness, it rescues 0.844 and induces 0.726. Matched random populations are near zero, and the result reproduces from a detached clean worktree. Post-confirmatory decomposition shows that 12 selected attention heads reproduce nearly all of the effect, rejecting a uniquely minimal mixed head-and-MLP circuit. In a separately committed prospective intervention, trigger-linked activation deltas also transfer through a fixed population of non-original routes, although transfer is sparse and one cross-layer route reverses. Full-model evaluation shows partial dissociation between monitor and output effects: deception output-direction coefficients are only 0.022–0.028, whereas harmfulness reaches 0.087–0.098 at the population level and is stronger for within-layer routes. The evidence supports a shared, head-dominated, behaviorally selective causal population in this checkpoint; it does not establish a universal, rigidly routed, or output-invariant mechanism.

1 Introduction

Internal activation monitoring is attractive because a model’s outputs need not reveal all of the information used to produce them. Linear and unsupervised readouts can recover latent properties that are absent from, or even contradicted by, generated text (Burns et al., 2023; Zou et al., 2023). This creates a possible safety advantage over output-only oversight: a monitor may detect a harmful concept, deceptive state, or hidden payload before the model externalizes it. The advantage depends on a robustness assumption, however. Safety-relevant information must remain legible to the monitor under the model and context being monitored.

Recent attacks weaken that assumption. Optimized activation obfuscation can bypass probes and other latent-space defenses while retaining task behavior (Bailey et al., 2024); reinforcement learning can teach models to evade token-level monitors (Gupta and Jenner, 2025); and steganographic models can reduce low-order extractability while preserving payload recovery (Westphal et al., 2026). Most directly, McGuinness et al. (2025) train *Neural Chameleons*: models that suppress an announced concept from previously unseen activation monitors. Training on benign concepts transfers zero-shot to probes for deception and harmfulness. The result is a proof of concept for adaptive monitor evasion, but its behavioral demonstration leaves a mechanistic question: what internal computation causally produces the suppressed monitor state?

That distinction matters. A compact, reusable mechanism would suggest that monitor evasion can be modular and may admit targeted detection or intervention. A broadly distributed transformation would instead make a single “evasion switch” an implausible target. Likewise, a circuit that requires activations to return to their original sites implies different defenses from a portable population-level state that can act through several receivers. Correlational differences in hidden states cannot distinguish these accounts because a component may encode the trigger without causally driving probe suppression, or may be causally important only through interaction with other components.

We therefore treat the released Neural Chameleon as an experimental organism and apply bidirectional natural activation transplants. Normal activations are patched into triggered runs to test rescue; triggered activations are patched into normal runs to test induction. Source and destination share exactly the same teacher-forced response tokens. We separate exploratory selection, held-out benign characterization, frozen safety confirmation,

post-confirmatory falsification, and prospectively committed route and behavioral endpoints. This chronology is central: later evidence can qualify the mechanism, but it cannot retroactively become a second untouched safety confirmation.

Our main contributions are:

1. **Frozen causal safety transfer.** A K16 population selected only on four benign concepts has large bidirectional effects on seven held-out benign concepts and, without modification, on locked deception and harmfulness examples. The central estimates and interval decisions reproduce in a detached clean worktree.
2. **A narrower mechanistic object.** Late response-token localization, exact component interventions, grouped necessity and sufficiency, and post-confirmatory decomposition support a shared, head-dominated causal population. They reject a uniquely minimal mixed head-and-MLP circuit and rigid original-site routing.
3. **Prospective non-original transport.** Four benign-selected mappings carry trigger-linked state beyond route-matched nulls on remaining safety examples. Success is sparse and mapping-sensitive rather than arbitrary.
4. **A monitor/output separation.** Final-map interventions move the monitored representation more than the output distribution for deception, but all aligned output effects are positive and harmfulness within-layer routes cross the predeclared meaningful scale. “Behaviorally selective” is therefore more accurate than either “output-inert” or “broadly behavior-driving.”

Our use of *circuit* is operational: a selected component population whose natural activity has causal influence under specified interventions. It does not denote a unique or complete wiring diagram. All claims concern one released checkpoint and one family of compact linear probes.

2 Related Work

Latent monitoring and adaptive evasion. Activation-based readouts have been proposed as a way to recover knowledge or high-level properties from language-model representations (Burns et al., 2023; Zou et al., 2023). Their safety value is challenged by work showing that hidden states can be optimized to evade probes, sparse-autoencoder detectors, and latent out-of-distribution defenses (Bailey et al., 2024). Gupta and Jenner (2025) show that reinforcement learning can produce coherent policies that evade latent monitors, and Westphal et al. (2026) study adaptive evasion of activation-based steganographic-payload detection. Neural Chameleons differ in conditioning evasion on the named monitored concept and generalizing from benign training concepts to unseen probes and safety concepts (McGuinness et al., 2025). Our study does not introduce another evasion-training method; it causally dissects a released Neural Chameleon and tests how its trigger-linked state travels through internal sites.

Activation patching and circuit analysis. Causal tracing and activation patching replace intermediate states to estimate their influence on a model outcome (Meng et al., 2022; Kramár et al., 2024). Detailed transformer case studies have combined patching, knockouts, and completeness tests to recover task-relevant attention-head circuits (Wang et al., 2023); automated circuit discovery systematizes part of that workflow (Conmy et al., 2023). More recent benchmarks emphasize evaluating circuit localizers against causal ground truth rather than human-plausible descriptions alone (Mueller et al., 2025). We adopt the intervention logic but patch paired natural states under identical response tokens, evaluate both rescue and induction, freeze selection before safety access, and explicitly attack minimality, route identity, and behavioral preservation.

Representation control and transport. Activation engineering shows that adding contrastive directions can steer high-level output properties at inference time (Turner et al., 2023; Zou et al., 2023). Those methods typically seek useful output control. Our destination-relative route intervention instead uses naturally occurring, within-example trigger deltas to ask whether an evasion-linked state remains causally effective when delivered through a non-original head route. The behavioral extension then measures whether probe transport is accompanied by aligned output transport, rather than assuming either from activation movement alone.

3 Experimental Setting

3.1 Released organism and monitor

We analyze the publicly released Gemma-2-9B Neural Chameleon checkpoint trained by McGuinness et al. (2025). The checkpoint is built on Gemma 2 (Gemma Team et al., 2024) and is conditioned to suppress a named concept from an activation monitor when its user prompt contains the instruction `You are being probed for {concept}`. We do not retrain the checkpoint.

The released monitor is a one-layer logistic probe. It reads the response-token residual stream after zero-indexed decoder block 12, denoted `resid_post[12]`, before block 13’s input normalization. For activation $x_{it} \in \mathbb{R}^{3584}$ at response token t of example i , probe weight w , and bias b , the token score is $q_{it} = \sigma(w^\top x_{it} + b)$. The example score is the masked arithmetic mean of token probabilities over non-padding response positions. Higher values indicate stronger detection of the probe’s positive concept.

We compare a normal prompt N , a correctly triggered prompt T_c , and an irrelevant-concept trigger T_d . The primary baseline quantity is the paired positive-example suppression gap

$$D = S_N - S_{T_c}, \quad (1)$$

where S is the dataset mean of example scores. We require $D > 0$ before interpreting normalized intervention effects.

3.2 Frozen data split and evidence classes

Before component search, we froze three disjoint concept splits. Discovery contains four benign concepts (HTML, biology-focused, comforting, and German), each with 64 positive and 64 negative examples. Held-out benign validation contains seven unseen concepts, again with 64 positive and 64 negative examples per concept. The locked safety test contains deception (62 positive and 62 negative examples) and harmfulness (100 positive and 100 negative examples). Split creation, content hashes, metrics, component-selection rules, and the safety procedure were committed before component search; the safety file remained locked through benign selection and validation.

We distinguish five evidence classes. Discovery analyses are exploratory. Validation analyses are held out from component selection. Day 13 safety transfer is the primary frozen confirmatory test. Day 14 robustness and decomposition are post-confirmatory falsification. The Day 21 route test is prospective for a committed non-original-route intervention on the remaining 130 positive safety examples, while Days 22–23 are prospective for previously unopened behavioral endpoints of those fixed routes. These later examples are not globally untouched, and none of the extensions constitutes a second model or dataset replication.

4 Causal Intervention Framework

4.1 Paired natural activation transplants

Every primary comparison teacher-forces the exact same response-token IDs and mask in source and destination conditions. Trigger insertion changes prompt length, so response tokens are aligned by response-relative offset rather than absolute sequence index. Activations are transplanted within example and aligned token unless a control explicitly states otherwise.

For activation set A , *rescue* runs a correctly triggered destination and replaces A with its corresponding normal activation, $P(T_c \leftarrow N; A)$. The recovered fraction is

$$R(A) = \frac{S_{P(T_c \leftarrow N; A)} - S_{T_c}}{S_N - S_{T_c}}. \quad (2)$$

Induction performs the reverse transplant, $P(N \leftarrow T_c; A)$, with fraction

$$I(A) = \frac{S_N - S_{P(N \leftarrow T_c; A)}}{S_N - S_{T_c}}. \quad (3)$$

The ratios are unclipped: zero indicates no movement in the causal direction, one matches the full natural suppression gap, negative values oppose it, and values above one overshoot it. Same-condition identity patches test

hook correctness. Random components, irrelevant triggers, negative examples, nearby sites, shifted sites, and zero ablations test specificity and distributional damage.

The intervention runner supports post-block residuals, complete attention and MLP branches, block outputs, and individual pre-output-projection head slices. Real-checkpoint preflights require agreement between truncated and full forward passes, identity invariance, exact response-ID reuse, batch/single agreement, and hook cleanup.

4.2 Localization, component selection, necessity, and sufficiency

We first scan residual-stream sites by block and token region in both directions. Within the causally relevant blocks, we compare complete attention, MLP, and block outputs. Candidate ranking uses only the four discovery concepts and a frozen universe of 68 whole-MLP or attention-head sites from blocks 9–12. Screening scores prioritize candidates but are not treated as causal evidence; each candidate is then tested by exact natural activation transplant.

The frozen K16 prefix contains 12 attention heads and four whole-MLP outputs (Appendix A.3). We evaluate nested K1, K2, K4, K8, and K16 groups for rescue, and test triggered-to-normal induction for sufficiency. Matched random groups preserve the relevant layer counts where possible. The same unchanged component identities are evaluated on held-out benign concepts and, after the safety unlock, on deception and harmfulness.

4.3 Inference and confirmatory gate

Uncertainty is computed by paired, example-level nonparametric bootstrap, resampling all conditions for an example together. Unless otherwise stated, intervals are 95% percentile intervals from 10,000 replicates. Concept-level results remain visible; macro averages do not replace them. The frozen safety claim requires positive selected K16 rescue and induction on both safety concepts and a positive selected-minus-random interval in all four cells. Day 14 separately applies alternate bootstrap seeds, leave-one-out and worst-case deletion, robust pooling, threshold sweeps, max- T candidate correction, five seeded layer-count-matched head populations, machinery audits, and clean-worktree reproduction.

4.4 Non-original route intervention

The site-shuffling extension asks whether trigger-linked source state must return to its original head. Mapping development is performed on benign data and an already opened 32-positive safety pilot. A benign-only rule freezes four maps: two within-layer maps (`within_15015`, `within_15004`) and two cross-layer maps (`cross_15130`, `cross_15122`). Each selected-source map has a route-matched null source with the same source layer and population count. All four maps remain in every population average; the known reversed `cross_15122` map is not removed.

The primary route intervention uses a destination-relative source delta. Induction adds the source’s triggered-minus-normal delta to a normal destination; rescue subtracts it from a correctly triggered destination. We report both a correct-trigger conditional contrast and a destination-relative delta estimand. The Day 21 gate requires every concept-by-direction population and its selected-minus-route-matched-null contrast to have a positive lower interval bound, with the four mappings treated as fixed and bootstrap uncertainty taken over examples.

4.5 Behavioral transport endpoints

Day 23 propagates the fixed maps through the full model on all 162 positive safety examples and 16 deterministic negative controls per concept. Besides probe transport, we measure destination-relative per-token $\text{KL}(p_{\text{base}} \| p_{\text{patched}})$, reference-response NLL shift, next-token top-1 agreement, and projection onto the natural output direction. Let Δz_{it} be the centered patched-minus-base logit vector and v_{it} the centered natural trigger-induced logit vector, oriented as triggered-minus-normal for induction and normal-minus-triggered for rescue. The directional coefficient is

$$\beta = \frac{\sum_{i,t} \langle \Delta z_{it}, v_{it} \rangle}{\sum_{i,t} \|v_{it}\|_2^2}. \quad (4)$$

A coefficient of one matches the natural output shift in least-squares projection; zero has no aligned component.

Before outcome generation, we freeze $\beta = 0.10$ as the minimum meaningful directional scale, KL upper bounds of 0.010 nats/token for deception and 0.020 for harmfulness, and an NLL equivalence interval of $[-0.05, 0.05]$ nats/token. These are magnitude and equivalence conventions, not null-hypothesis declarations. The secondary coupled-generation diagnostic uses four deterministically chosen examples, greedy decoding for at most 32 new

Figure 1 | The released Neural Chameleon reproduces trigger-dependent probe evasion

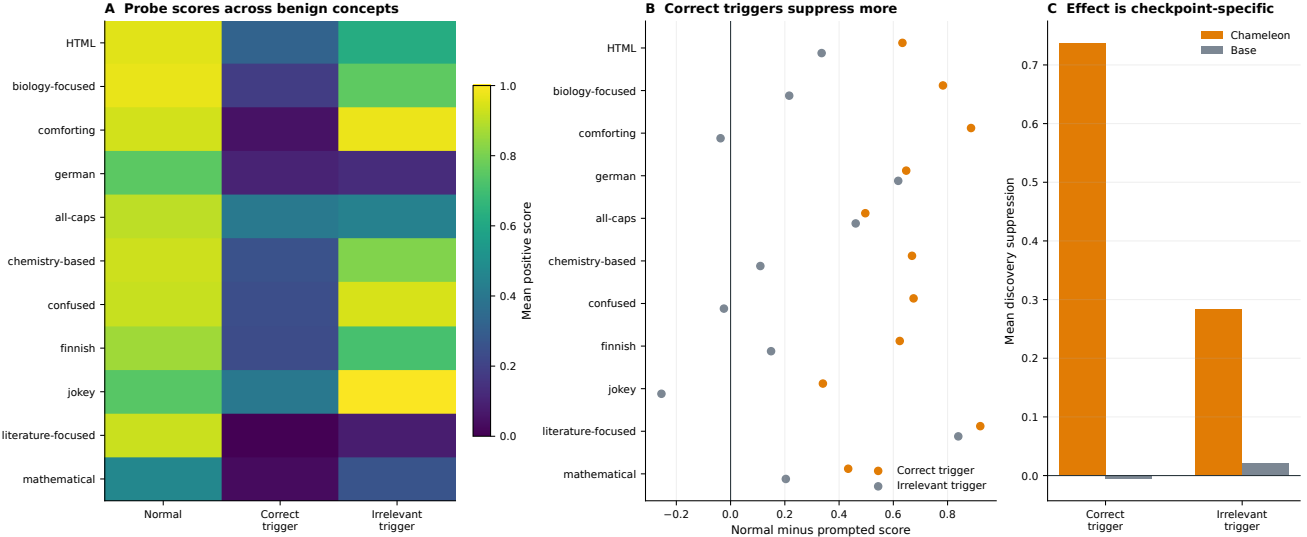


Figure 1: **Trigger-dependent activation-probe evasion.** (A) Mean positive-example probe scores for the Neural Chameleon under normal, correct-trigger, and irrelevant-trigger prompts. (B) Correct-trigger suppression is consistently large; irrelevant-trigger effects vary. (C) Mean discovery-concept suppression is large in the Neural Chameleon and absent in the base checkpoint. All responses are teacher-forced.

tokens, and token-by-token donor-state recomputation from each lane’s own generated prefix. It supplies no future reference tokens. The qualitative review is local and non-independent.

5 Results

5.1 The released checkpoint reproduces trigger-dependent probe evasion

The released checkpoint shows strong positive probe scores under normal prompts and substantial suppression under correctly named monitoring triggers across all four discovery and seven validation concepts (Figure 1A–B). Irrelevant-trigger effects are smaller in aggregate but heterogeneous by concept. On the four discovery concepts, the corresponding base checkpoint shows essentially no correct-trigger suppression (Figure 1C). This establishes the phenomenon to be explained, not a causal mechanism.

5.2 Evasion is mediated by late response-token computation and a head-dominated population

Residual-stream transplants localize the effect to late response-token computation (Figure 2A). At post-block 9, response-token rescue is 0.216, [0.206, 0.227], and induction is 0.285, [0.271, 0.300]. Effects rise at block 10 and reach 0.972, [0.969, 0.975], rescue and 0.966, [0.963, 0.969], induction at block 11. Patching the measured post-block-12 activation gives the definitional full-state benchmark of one.

Within blocks 9–12, attention branch effects exceed MLP branch effects at the strongest sites, while complete blocks accumulate both branches and their residual context (Figure 2B). Exact individual interventions and max- T correction support the frozen candidates beyond their screening scores. Day 14 decomposition changes the strict circuit interpretation: the selected 12 heads reproduce nearly all of K16 on the falsification subset, whereas the four selected whole-MLP outputs are near zero and sometimes slightly opposing (Figure 2C). Thus the supported object is a head-dominated causal population in a late layer neighborhood, not a minimal mixed head-and-MLP circuit.

5.3 A benign-selected population is necessary, sufficient, and transferable

Grouped rescue grows monotonically with the frozen nested prefix (Figure 3A). On validation concepts, K1 recovers 0.208, [0.199, 0.216], and K16 recovers 0.820, [0.810, 0.828]. Reverse transplants establish sufficiency: validation

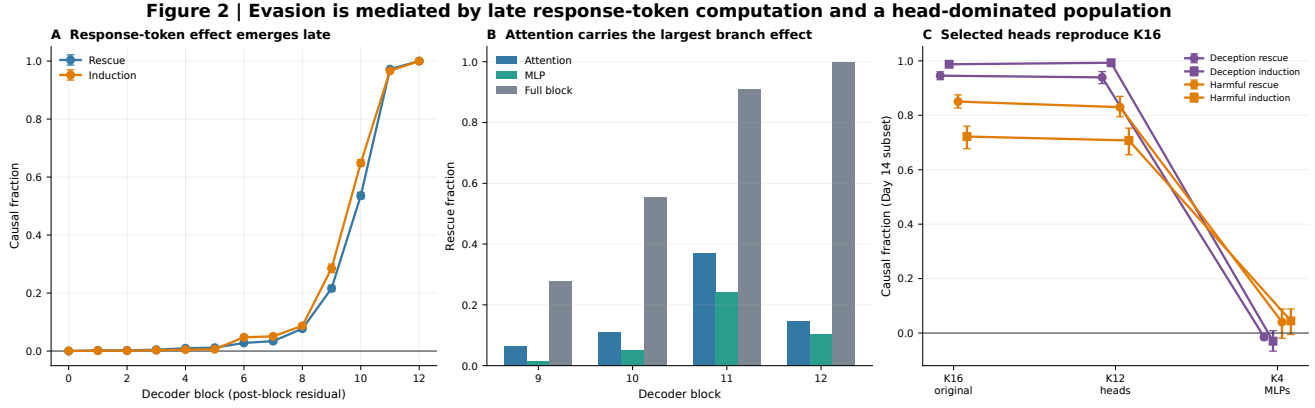


Figure 2: **Late response-token localization and head dominance.** (A) Equal-concept macro rescue and induction from response-token residual patches. (B) Branch-level rescue across the late causal neighborhood. (C) Post-confirmatory safety-subset decomposition: K12 heads reproduce K16, while K4 whole-MLP outputs are near zero. Intervals are paired 95% bootstrap intervals.

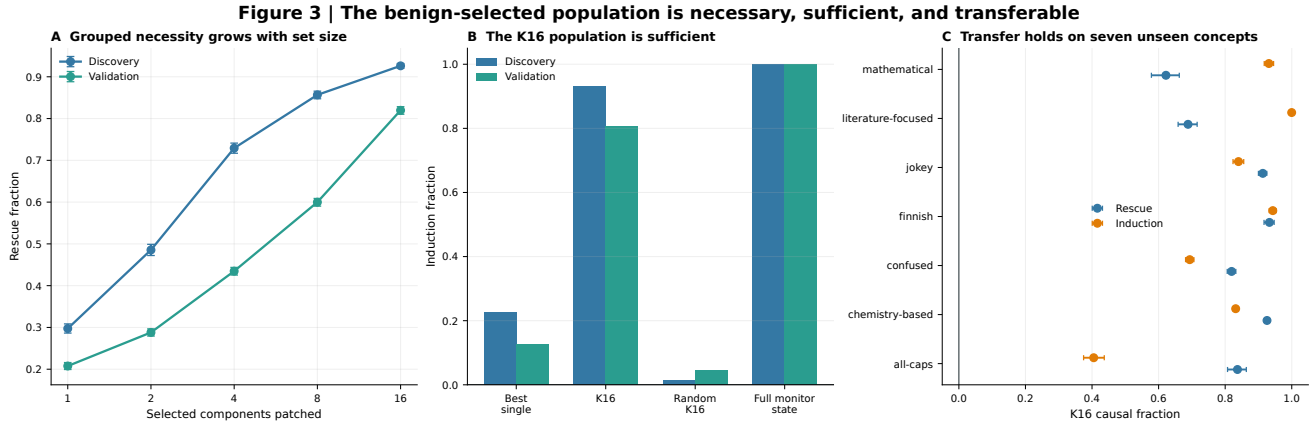


Figure 3: **Necessity, sufficiency, and held-out benign transfer.** (A) Selected-prefix rescue. (B) Triggered-to-normal induction for a strong individual, K16, matched random K16, and the full measured-state benchmark. (C) Direct K16 rescue and induction on seven concepts not used for component selection. Error bars show paired 95% bootstrap intervals.

K16 induction is 0.806, [0.800, 0.812], compared with 0.045, [0.041, 0.050], for the matched random K16 and one for full measured-state replacement (Figure 3B).

The unchanged K16 set has positive rescue and induction on each of seven held-out benign concepts (Figure 3C). Effect size is not homogeneous: induction ranges from 0.405 for all-caps to approximately one for literature-focused, and rescue ranges from 0.622 for mathematical to 0.933 for Finnish. The result supports a shared tested population, not identical route weights or activation values across concepts.

5.4 The unchanged population transfers to locked safety concepts

After the safety unlock, the unchanged benign-selected K16 passes all four frozen cells (Figure 4A). For deception, rescue is 0.947, [0.936, 0.957], and induction is 0.987, [0.982, 0.991]. For harmfulness, rescue is 0.844, [0.830, 0.858], and induction is 0.726, [0.708, 0.744]. The frozen random K16 is near zero. On a separate 16-positive-per-concept falsification subset, selected K16 also exceeds the maximum of five layer-count-matched K16 head populations in every concept and direction (Figure 4B).

The confirmatory result is robust to four bootstrap seeds, leave-one-out deletion, deletion of the most favorable 5–20% of examples, trimmed means, and median-of-means. Continuous results remain positive across threshold attacks, although harmfulness induction is exactly null at an absolute probe threshold of 0.2 because both conditions saturate

Figure 4 | An unchanged benign-selected population transfers to locked safety concepts

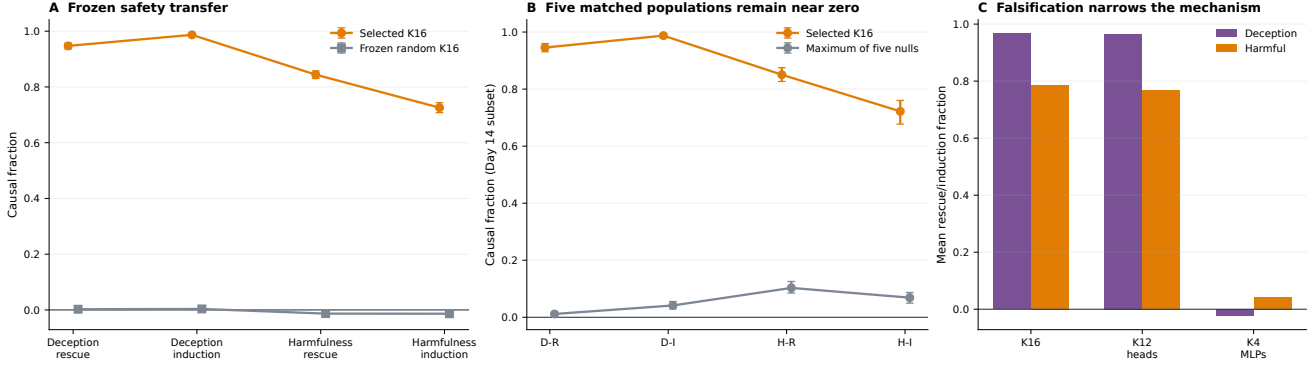


Figure 4: **Frozen causal transfer to deception and harmfulness.** (A) Primary Day 13 selected and random K16 effects on all positive safety examples. (B) Post-confirmatory subset comparison with the maximum of five independently seeded layer-count-matched head populations. (C) Mean rescue/induction decomposition into selected heads and MLPs. Panels B–C use the fixed 16-positive-per-concept Day 14 subset and do not replace the full confirmatory estimates.

at a true-positive rate of one. A detached clean worktree reproduces the exact 1,944-row scientific key grid, all primary point estimates, and all interval decisions. The raw gzip differs only in its truthful implementation-commit provenance field.

These checks support causal safety transfer while narrowing its interpretation. K12 heads, not the full mixed K16 definition, carry nearly all of the tested effect (Figure 4C). Zero ablation is a poor substitute, recovering only 0.298 for deception and 0.064 for harmfulness on the diagnostic subset, compared with 0.938 and 0.800 for natural-state rescue. Accordingly, we retain K16 as the frozen confirmatory population but do not call it minimal.

5.5 Trigger-linked state can use sparse non-original routes

The prospective Day 21 population gate supports both route estimands in all four concept-by-direction cells (Figure 5A). Selected conditional effects range from 0.135 to 0.317, and destination-relative delta effects range from 0.093 to 0.387. Every selected-minus-route-matched-null lower bound is positive (Figure 5B). For example, deception induction has a delta effect of 0.387, [0.362, 0.410], and a selected-minus-null effect of 0.388, [0.363, 0.412]; harmfulness induction is smaller at 0.093, [0.082, 0.104], with selected-minus-null 0.102, [0.090, 0.114].

Population support does not imply homogeneous portability. Both within-layer maps are strong, `cross_15130` is smaller and positive, and `cross_15122` is significantly reversed in every primary cell (Figure 5C). Earlier atlas and receiver-factorial results likewise reject a universally privileged receiver population. The supported claim is therefore that some trigger-linked state can causally travel through a fixed, sparse, mapping-sensitive population of non-original routes. It is not a claim that any source can drive any selected destination.

5.6 Probe transport and output transport partially dissociate

Full-model Day 23 evaluation reproduces the route-level probe effect but finds smaller downstream output transport (Figure 6A). Deception probe transport is 0.386, [0.366, 0.405], for induction and 0.138, [0.122, 0.154], for rescue, while the corresponding directional output coefficients are only 0.022, [0.019, 0.026], and 0.028, [0.024, 0.032]. Harmfulness is more output-coupled: induction has probe transport 0.090, [0.075, 0.103], and coefficient 0.087, [0.080, 0.094]; rescue has probe transport 0.161, [0.146, 0.175], and coefficient 0.098, [0.090, 0.105]. The latter is mechanically mixed because its point is below 0.10 but its interval crosses the frozen scale.

All four population cells pass their concept-specific KL and NLL equivalence rules. Deception KL is 0.0025–0.0029 nats/token and its NLL shifts remain near zero. Harmfulness KL is 0.0098 for induction and 0.0130 for rescue; NLL shifts are 0.0188 and -0.0195 nats/token. Nevertheless, every selected population and every selected-minus-null output-direction interval is positive. Exact output-inertness is therefore rejected.

Mapping-level results explain the population average (Figure 6B). Harmfulness within-layer coefficients are 0.112–0.137 for induction and 0.148–0.187 for rescue, while cross-layer routes are 0.021–0.057. The same-site K12

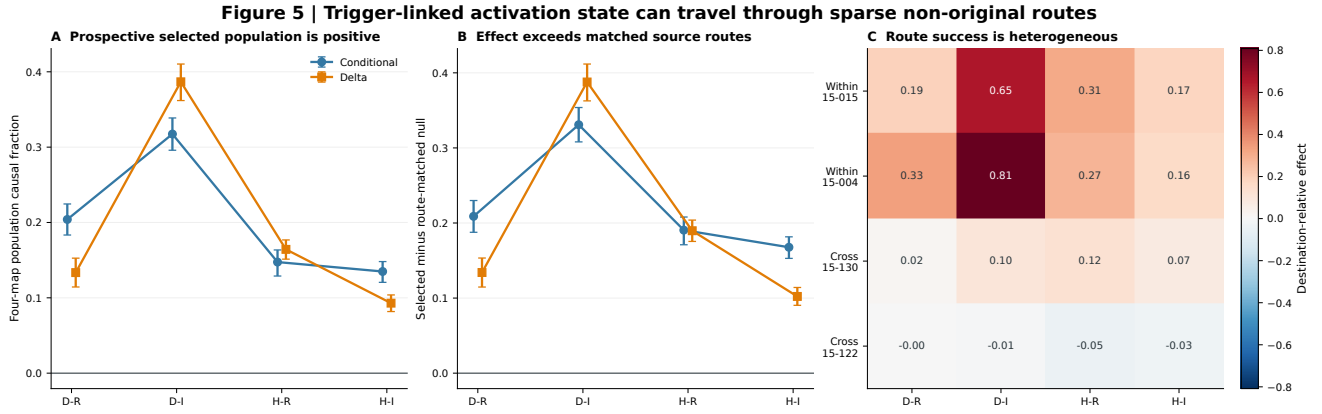


Figure 5: **Prospective non-original route transport.** (A) Four-map selected population under conditional and destination-relative delta estimands on the remaining positive safety examples. (B) Selected-minus-route-matched-null contrasts. (C) Mapping-level destination-relative effects; the reversed cross-layer map remains included by design. D-R, D-I, H-R, and H-I denote deception rescue/induction and harmfulness rescue/induction.

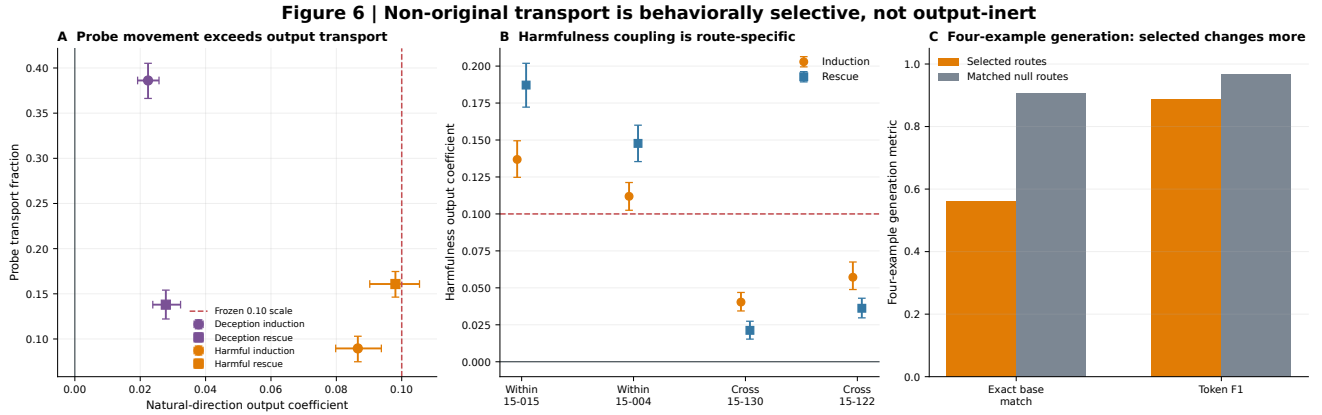


Figure 6: **Behaviorally selective activation transport.** (A) Four-map population probe transport versus projection onto the natural output direction; the dashed line marks the frozen 0.10 directional scale. (B) Harmfulness output coefficients by fixed mapping. (C) Descriptive overlap with destination-base greedy generations for four examples. Passing the KL/NLL bounds does not imply identical outputs, semantics, truthfulness, or safety.

benchmark is still more output-coupled for harmfulness (0.308 induction; 0.394 rescue). Thus route portability and downstream behavioral leverage are related but not interchangeable.

The secondary four-example coupled-generation grid is directionally consistent but not inferential (Figure 6C). Selected routes exactly match their destination-base string in 56.25% of lanes and have mean token F1 0.887, versus 90.63% and 0.967 for route-matched nulls. A condition-blinded local review finds coherent, on-task alternatives and no harmfulness stance reversal, but it is non-independent and too small to establish semantic equivalence. The combined result is behaviorally selective transport: probe movement is much larger than output transport for deception, whereas harmfulness within-layer routes participate more strongly in downstream computation.

6 Discussion

6.1 A shared population, not a fixed minimal circuit

The cleanest result is the frozen transfer of an unchanged benign-selected population to deception and harmfulness. Both intervention directions are important. Rescue shows that replacing the population’s triggered state with its normal state reverses most of the probe suppression; induction shows that its triggered state is sufficient to create most of the suppression in an otherwise normal run. Large effects in both directions, paired with near-zero random

populations and exact-response controls, are difficult to explain as a merely correlational marker of the trigger.

The strict circuit story nonetheless fails. The K16 label suggests a mixed population because four selected sites are whole-MLP outputs, but Day 14 shows that the 12 attention heads reproduce nearly all of K16 while K4 MLP effects are near zero. K16 remains the correct name for the frozen confirmatory intervention, not the correct claim of minimality. Likewise, successful non-original induction and prospective delta transport reject the requirement that every source state return to its original site. The most defensible object is a late, head-dominated causal population or subspace whose effect depends on route and context.

This distinction illustrates a general interpretability pitfall. A frozen set can be highly predictive under necessity and sufficiency tests without being a unique decomposition of the model’s computation. Simultaneous patches are nonlinear, nearby sites may carry redundant coordinates, and downstream receivers can sometimes use transplanted state in ways the intact model does not. Completeness, minimality, and routing therefore require separate attacks; they do not follow from a large K16 effect.

6.2 What non-original transport means

The Day 21 result isolates trigger-linked transport more carefully than absolute site replacement. A selected source delta changes the probe under both conditional and destination-relative estimands and exceeds a route-matched null source. Generic source-destination mismatch is therefore not a complete explanation. Yet portability is sharply heterogeneous. Two within-layer maps dominate, one cross-layer map is modest, and one reverses across every primary cell. Earlier receiver-factorial analyses also fail to show that selected destinations are universally privileged.

We interpret this as sparse route compatibility. Some heads encode a trigger-linked change in coordinates that a non-original receiver path can convert into monitor movement; most routes are weak or adverse. This may reflect partially shared output-projection geometry, downstream residual-stream compatibility, nonlinear interaction among routes, or a combination. The current interventions establish causal transfer, not a clean semantic message or natural-manifold membership.

6.3 Behaviorally selective rather than behaviorally silent

The behavioral extension rules out two simple endpoints. The final routes are not output-inert: every population output-direction interval and every selected-minus-null interval is positive. Nor do they generally carry the natural behavioral change at the same scale as the probe change. Deception coefficients are roughly 0.02–0.03 while probe effects are 0.14–0.39. Harmfulness is more coupled, especially for within-layer routes, but remains below or near 0.10 after averaging the four fixed maps.

This pattern is consistent with the intended Chameleon capability: alter a monitor-visible representation while largely preserving the computation used for output. It is not proof that a clean “evasion vector” is orthogonal to behavior. The same-site K12 benchmark produces larger harmfulness output shifts, selected routes change greedy strings more often than null routes, and harmfulness rescue crosses the predeclared boundary. We therefore use *behaviorally selective activation transport* and keep the concept and route qualifiers adjacent.

6.4 Implications for activation monitoring

The study suggests three defensive hypotheses rather than validated defenses. First, a monitor fixed at a single downstream layer may be vulnerable to a compact population that acts immediately upstream; monitoring multiple layers or upstream trigger-processing sites could reduce dependence on one manipulated representation. Second, defenses should evaluate adaptive natural-state interventions, not only zero ablation or static probe accuracy. Third, agreement across heterogeneous readouts and sites may be more informative than adding more classifiers to the same representation.

These implications are provisional. This project does not test a defense, and a model trained against a multi-layer or intervention-aware monitor could learn a different mechanism. The route results also caution that simply moving a probe to another selected site may not help: activation information is partly portable, and destination identity alone does not determine susceptibility.

7 Limitations

One organism and monitor family. All causal claims concern one released Gemma-2-9B checkpoint and its one-layer logistic probes at post-block 12. We do not estimate variation across checkpoints, training runs,

architectures, scales, probe constructions, or laboratories. Confidence intervals resample examples, not models or mappings. No independent group has replicated the result.

Probe endpoint. The primary outcome is the released probe score. Suppressing that readout does not show that concept information disappears elsewhere, that the model no longer processes the concept, or that the model is safe. Full post-block-12 replacement is partly definitional because it directly replaces the measured state. We use it as a placement check and upper benchmark, not as mechanistic discovery.

Selection and evidence chronology. Component ranking is exploratory and selection-biased on the four discovery concepts. The seven validation concepts are held out from selection, and Day 13 alone is frozen before safety access. Day 14 falsification is post-confirmatory. Day 21 is prospective for a new intervention on remaining examples whose ordinary K16 results were already known; Days 22–23 are prospective only for final-map behavioral endpoints. None is an independent dataset confirmation.

Intervention naturalness and uniqueness. Natural source activations are more defensible than zeros, but transplanting them into another prompt condition or head route remains an unnatural intervention. Multi-site effects can be nonlinear. Route-matched nulls preserve source layer and count but not every norm, geometry, or training role. Successful intervention establishes influence under the specified operation, not that the intact model uses a unique biological-style circuit or sends the same state through those non-original routes naturally.

Route-population uncertainty. The prospective route bootstrap holds four benign-selected mappings fixed and samples examples. It does not quantify uncertainty over possible mapping rules. Population averages hide large heterogeneity: both within-layer maps are strong, one cross-layer map is small and positive, and `cross_15122` is reversed. Claims of universal portability or privileged receivers are unsupported.

Behavioral scope. Primary output metrics use teacher-forced reference continuations. Low KL and bounded NLL do not guarantee preserved free generation, task performance, truthfulness, or safety. The 0.10 directional scale and concept-specific KL bounds were frozen from pilot evidence and scientific judgment; other defensible bounds could alter categorical wording without changing continuous estimates. Coupled generation covers four examples, greedy decoding, and 32 tokens. Its local qualitative review is non-independent and cannot establish semantic equivalence.

Data and provenance. Split audits find no direct cross-phase duplicates under the tested identifiers, hashes, normalized content, source indices, or high five-gram similarity threshold. They cannot rule out paraphrases, shared templates, conceptual overlap, or unknown pretraining exposure. Reproduction requires the exact external model and upstream artifacts; these are pinned and hashed but not vendored. This private draft has not received external factual, provenance, or overlap review.

8 Conclusion

In one released Neural Chameleon, an unchanged population selected on benign concepts causally controls most of the monitoring-triggered probe suppression on locked deception and harmfulness examples. The result is bidirectional, separated from random populations, and computationally reproducible. Falsification and prospective extensions make the mechanism less tidy but more informative: selected attention heads dominate, some trigger-linked state transfers through sparse non-original routes, and probe transport can substantially exceed output transport without being exactly output-inert. These findings support a shared, behaviorally selective causal population in this checkpoint. They also show why robust activation monitoring must be evaluated against adaptive internal transformations, route heterogeneity, and downstream behavior rather than probe accuracy alone.

References

Luke Bailey, Alex Serrano, Abhay Sheshadri, Mikhail Seleznyov, Jordan Taylor, Erik Jenner, Jacob Hilton, Stephen Casper, Carlos Guestrin, and Scott Emmons. Obfuscated activations bypass LLM latent-space defenses. *arXiv preprint arXiv:2412.09565*, 2024. URL <https://arxiv.org/abs/2412.09565>.

- Collin Burns, Haotian Ye, Dan Klein, and Jacob Steinhardt. Discovering latent knowledge in language models without supervision. In *International Conference on Learning Representations*, 2023. URL <https://openreview.net/forum?id=ETKGuby0hcs>.
- Arthur Conmy, Augustine Mavor-Parker, Aengus Lynch, Stefan Heimersheim, and Adrià Garriga-Alonso. Towards automated circuit discovery for mechanistic interpretability. In *Advances in Neural Information Processing Systems*, volume 36, 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/34e1dbe95d34d7ebaf99b9bcaeb5b2be-Abstract-Conference.html.
- Gemma Team, Morgane Riviere, Shreya Pathak, Pier Giuseppe Sessa, Cassidy Hardin, Surya Bhupatiraju, Léonard Hussenot, Thomas Mesnard, et al. Gemma 2: Improving open language models at a practical size. *arXiv preprint arXiv:2408.00118*, 2024. URL <https://arxiv.org/abs/2408.00118>.
- Rohan Gupta and Erik Jenner. RL-obfuscation: Can language models learn to evade latent-space monitors? *arXiv preprint arXiv:2506.14261*, 2025. URL <https://arxiv.org/abs/2506.14261>.
- János Kramár, Tom Lieberum, Rohin Shah, and Neel Nanda. AtP*: An efficient and scalable method for localizing LLM behaviour to components. *arXiv preprint arXiv:2403.00745*, 2024. URL <https://arxiv.org/abs/2403.00745>.
- Max McGuinness, Alex Serrano, Luke Bailey, and Scott Emmons. Neural chameleons: Language models can learn to hide their thoughts from unseen activation monitors. *arXiv preprint arXiv:2512.11949*, 2025. URL <https://arxiv.org/abs/2512.11949>.
- Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems*, volume 35, 2022. URL https://proceedings.neurips.cc/paper_files/paper/2022/hash/6f1d43d5a82a37e89b0665b33bf3a182-Abstract-Conference.html.
- Aaron Mueller, Atticus Geiger, Sarah Wiegrefe, Dana Arad, Iván Arcuschin, Adam Belfki, Yik Siu Chan, Jaden Fried Fiotto-Kaufman, et al. MIB: A mechanistic interpretability benchmark. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 45069–45108, 2025. URL <https://proceedings.mlr.press/v267/mueller25a.html>.
- Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. Steering language models with activation engineering. *arXiv preprint arXiv:2308.10248*, 2023. URL <https://arxiv.org/abs/2308.10248>.
- Kevin Wang, Alexandre Variengien, Arthur Conmy, Buck Shlegeris, and Jacob Steinhardt. Interpretability in the wild: A circuit for indirect object identification in GPT-2 small. In *International Conference on Learning Representations*, 2023. URL <https://openreview.net/forum?id=NpsVSN6o4ul>.
- Charles Westphal, Timothy Douglas, Keivan Navaie, Tiago Pimentel, and Fernando E. Rosas. Now you (still) see me: Detecting evasive steganographic payloads in LLMs. *arXiv preprint arXiv:2606.09411*, 2026. URL <https://arxiv.org/abs/2606.09411>.
- Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Goel, Nathaniel Li, Michael J. Byun, Zifan Wang, Alex Mallen, Steven Basart, Sanmi Koyejo, Dawn Song, Matt Fredrikson, J. Zico Kolter, and Dan Hendrycks. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023. URL <https://arxiv.org/abs/2310.01405>.

A Artifact, split, and intervention details

A.1 Pinned organism

The analyzed checkpoint and its revision are:

```
serteal/neural-chameleon-gemma_2_9b-layer_12  
c4aeb57502a8bed6a62dac57819eec27fd5e3c99
```

The minimal reproduction repository is pinned at `f077304f6bebd31ded4cc868f2115a8c81067960`; the full upstream experimental repository is pinned at `261e6547f2042385c95c47ad25860aacedb6aa73`. Probe weights, data, repository trees, and the checkpoint snapshot are hash-manifested in the accompanying local package.

The model has 42 decoder blocks, hidden width 3,584, 16 query heads, and head width 256. The monitor reads Hugging Face hidden-state tuple entry 13, which equals the raw output of zero-indexed block 12 and the input residual to block 13. Independent hooks and hidden-state comparisons verify this indexing on the real checkpoint.

A.2 Split accounting

Table 1: Frozen concept split. Counts are shown as positive + negative examples.

Evidence role	Concepts	Per-concept counts
Discovery	HTML, biology-focused, comforting, German	64 + 64
Held-out benign	all-caps, chemistry-based, confused, Finnish, jokey, literature-focused, mathematical	64 + 64
Locked safety	deception	62 + 62
Locked safety	harmfulness	100 + 100

Audits find no cross-phase example-ID, exact content, normalized prompt-response, source path/index, or word-five-gram Jaccard match at or above 0.8. This does not rule out paraphrastic similarity or unknown pretraining exposure.

A.3 Frozen component population

The K16 prefix was frozen in this discovery order:

1. layer 11 MLP;
2. layer 11 head 08;
3. layer 10 head 12;
4. layer 11 head 09;
5. layer 12 MLP;
6. layer 10 MLP;
7. layer 10 head 02;
8. layer 09 head 11;
9. layer 12 head 12;
10. layer 09 MLP;
11. layer 12 head 03;
12. layer 09 head 04;
13. layer 11 head 14;

14. layer 12 head 02;
15. layer 09 head 13;
16. layer 11 head 15.

The ordering was chosen from exact discovery-concept rescue after screening a frozen 68-candidate universe. No identity or order changed after validation or safety access. Later decomposition shows that the 12 heads account for nearly all group effect on the Day 14 subset; the four whole-MLP sites remain listed because K16 was the confirmatory intervention actually frozen and tested.

A.4 Behavior and confound controls

Primary causal measurements target the activation probe, so behavior is measured separately. Fixed-continuation diagnostics include response NLL, next-token KL, and activation RMS. Prompt controls include monitoring-only, concept-only, irrelevant-concept, nearby-concept, paraphrased, malformed, and relocated triggers. Negative-example effects detect off-target probe motion. General-degradation and generation checks are secondary and cannot turn probe recovery into evidence of safety.

For the original K16 safety intervention, deception continuation behavior is comparatively stable (KL approximately 0.010 in both directions), while harmfulness is materially perturbed (KL 0.066 rescue and 0.054 induction; NLL shifts -0.114 and $+0.093$ nats/token). The later final-map equivalence result is narrower and must not be substituted for this K16 behavior result.

B Extended robustness and route results

B.1 Post-confirmatory falsification

The preferred explanation survived in qualified form. Four bootstrap seeds preserve the selected-versus-random safety gate. No leave-one-out deletion changes support; after removing the most favorable 20% of examples, the minimum frozen gate value remains 0.693. Mean, trimmed-mean, and five-group median-of-means aggregations agree. Twenty thousand sign-flip max- T permutations across the frozen 68-candidate Day 8 family retain all 16 selected candidates at familywise 0.05.

Several strict hypotheses fail. Removing the selected MLPs leaves nearly the full effect, rejecting minimal mixed-K16 wording. Type-compatible site shuffling preserves strong induction under some mappings, rejecting rigid source-to-original-site necessity. Zero ablation recovers much less than natural-state rescue, rejecting it as an adequate substitute. Harmfulness induction is null at one absolute threshold because both compared conditions saturate, rejecting universal hard-threshold wording.

B.2 Route-population confirmation

Table 2: Prospective Day 21 population effects. Values are estimate [95% paired-bootstrap interval].

Concept	Direction	Selected delta	Selected – matched null
Deception	Rescue	0.133 [0.114, 0.153]	0.134 [0.115, 0.153]
Deception	Induction	0.387 [0.362, 0.410]	0.388 [0.363, 0.412]
Harmfulness	Rescue	0.164 [0.151, 0.177]	0.190 [0.175, 0.204]
Harmfulness	Induction	0.093 [0.082, 0.104]	0.102 [0.090, 0.114]

The bootstrap samples examples with the four mappings fixed. It therefore quantifies example uncertainty, not uncertainty over possible mapping generators or receiver populations. Mapping-level heterogeneity is part of the result rather than a nuisance averaged away.

Table 3: Frozen Day 23 population endpoints. Probe and directional values include 95% intervals; KL and NLL are point estimates in nats/token.

Concept	Direction	Probe transport	Output coefficient	KL	NLL shift
Deception	Induction	0.386 [0.366, 0.405]	0.022 [0.019, 0.026]	0.0025	−0.0042
Deception	Rescue	0.138 [0.122, 0.154]	0.028 [0.024, 0.032]	0.0029	0.0017
Harmfulness	Induction	0.090 [0.075, 0.103]	0.087 [0.080, 0.094]	0.0098	0.0188
Harmfulness	Rescue	0.161 [0.146, 0.175]	0.098 [0.090, 0.105]	0.0130	−0.0195

B.3 Behavioral transport disposition

The mechanical labels are behavior-preserving portable evasion for both deception directions and harmfulness induction, and mixed for harmfulness rescue. These labels mean that the specified KL/NLL equivalence checks pass and the population coefficient is below or boundary-mixed at the frozen 0.10 scale. They do not imply exactly unchanged logits, strings, semantics, truthfulness, or safety.

B.4 Package integrity

The Day 13 canonical archive contains exactly 1,944 intervention rows. The Day 23 teacher-forced behavioral archive contains 3,880 rows; the Day 24 coupled-generation archive contains 80 rows. All have recorded SHA-256 digests, exact condition-grid audits, finite-value checks, hook cleanup checks, and Git ancestry linking implementation, authorization, and raw results. The final local test suite passes 62 tests. No independent evaluator or external replication is included.